

Практика SPARK

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Группа : Управление на основе данных

Содержание отчета:

1. Colab Notebook с выполненной работой
ссылка на colab →
(https://colab.research.google.com/drive/1J0NPPDdBsk2MPxRrMyKE4i_H4sqiI9ws?usp=sharing)
2. Результаты выполнения скрипта в Minio S3

Подключим Spark

```
1 !pip install pyspark==3.4.1
2 from pyspark.sql import SparkSession
3 import sys
```

Requirement already satisfied: pyspark==3.4.1 in /usr/local/lib/python3.12/dist-packages (3.4.1)

Requirement already satisfied: py4j==0.10.9.7 in /usr/local/lib/python3.12/dist-packages (from pyspark==3.4.1) (0.10.9.7)

Инициализируем ключи доступа к S3

```
1 from pyspark.sql import SparkSession
2
3 ACCESS_KEY = "lab-1-user-7"
4 SECRET_KEY = "FIq9Q160hYaeYjvbAvAv"
5 ENDPOINT = "http://minio-1.devops-playground.innopolis.university"
6 BUCKET = "lab-1-bucket-1"
7 YOUR_NAME = "kirill-gusev"
```


Создание SparkSession

```
1 spark = SparkSession.builder \
2     .appName("Spark Lab") \
3     .master("local[*]") \
4     .config("spark.hadoop.fs.s3a.access.key", ACCESS_KEY) \
5     .config("spark.hadoop.fs.s3a.secret.key", SECRET_KEY) \
6     .config("spark.hadoop.fs.s3a.endpoint", ENDPOINT) \
7     .config("spark.hadoop.fs.s3a.path.style.access", "true") \
8     .config("spark.hadoop.fs.s3a.connection.ssl.enabled", "false") \
9     .config("spark.hadoop.fs.s3a.impl", "org.apache.hadoop.fs.s3a.S3AFileSystem") \
10    .config("spark.jars.packages", "org.apache.hadoop:hadoop-aws:3.3.4") \
11    .getOrCreate()
12
13 sc = spark.sparkContext
14 print("SparkSession создана")
```

SparkSession создана

Parallelize и coalesce

```
1 # Создание RDD
2 data = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15]
3 rdd = sc.parallelize(data, 5)
4 print(f"Исходное количество партиций: {rdd.getNumPartitions()}")
5
6 # Сохранение с 5 партициями
7 !rm -rf /content/rdd_output_5parts
8 rdd.saveAsTextFile("/content/rdd_output_5parts")
9 print("✅ Сохранено с 5 партициями")
10
11 # Уменьшение партиций
12 rdd_coalesced = rdd.coalesce(2)
13 print(f"После coalesce: {rdd_coalesced.getNumPartitions()}")
14
15 # Сохранение с 2 партициями
16 !rm -rf /content/rdd_output_2parts
17 rdd_coalesced.saveAsTextFile("/content/rdd_output_2parts")
18 print("✅ Сохранено с 2 партициями")
```

Исходное количество партиций: 5

✅ Сохранено с 5 партициями

После coalesce: 2

✅ Сохранено с 2 партициями

Загрузим нужные библиотеки для чтения файлов с S3 + подключимся к хранилищу

```
1 !pip install boto3 pandas
```

```
Requirement already satisfied: boto3 in /usr/local/lib/python3.12/dist-packages (1.42.68)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: botocore<1.43.0, ≥1.42.68 in /usr/local/lib/python3.12/dist-packages (from boto3) (1.42.68)
Requirement already satisfied: jmespath<2.0.0, ≥0.7.1 in /usr/local/lib/python3.12/dist-packages (from boto3) (1.1.0)
Requirement already satisfied: s3transfer<0.17.0, ≥0.16.0 in /usr/local/lib/python3.12/dist-packages (from boto3) (0.16.0)
Requirement already satisfied: numpy ≥1.26.0 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.0.2)
Requirement already satisfied: python-dateutil ≥2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz ≥2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata ≥2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.3)
Requirement already satisfied: urllib3≠2.2.0, <3, ≥1.25.4 in /usr/local/lib/python3.12/dist-packages (from botocore<1.43.0, ≥1.42.68→boto3) (2.5.0)
Requirement already satisfied: six ≥1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil ≥2.8.2→pandas) (1.17.0)
```

```
1 import boto3
```

```
1 import pandas as pd
```

```
1 from io import StringIO, BytesIO
```

Добавим модуль `boto3` для исправления технической ошибки подключения к S3 (после подключения к Minio конвертируем в `spark.createDataFrame()`)

Настроим подключение к S3

```
1 endpoint = 'https://minio-1.devops-playground.innopolis.university'
2 access_key = 'lab-1-user-7'
3 secret_key = 'FIq9Q160hYaeYjvbAvAv'
```

Прочитаем файл

```
1 bucket_name = 'lab-1-bucket-1'
2 object_key = 'kirill-gusev/retail/Online_Retail-1.csv'
```

```
1 session = boto3.session.Session()
2
3 s3_client = session.client(
4
5     service_name='s3',
6     endpoint_url='https://minio-1.devops-playground.innopolis.university',
7     aws_access_key_id='lab-1-user-7',
8     aws_secret_access_key='FIq9Q160hYaeYjvbAvAv',
9     verify=False
10
11 )
```

```
1 response = s3_client.get_object(Bucket='lab-1-bucket-1', Key='kirill-gusev/retail/Online_Retail-1.csv')
2 content = response['Body'].read().decode('utf-8')
3 df_pd = pd.read_csv(StringIO(content))
```

```
/usr/local/lib/python3.12/dist-packages/urllib3/connectionpool.py:1097: InsecureRequestWarning: Unverified HTTPS request is being made to host 'minio-1.devops-playground.innopolis.university'. Adding certifi warnings.warn(
```

Конвертируем в `spark.createDataFrame`

```
1 df_spark = spark.createDataFrame(df_pd)
```

```
1 df_spark.show(8)
```

InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
536365	85123A	WHITE HANGING HEA...	6	2010-12-01 08:26:00	2.55	17850	United Kingdom
536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	17850	United Kingdom
536365	84406B	CREAM CUPID HEART...	8	2010-12-01 08:26:00	2.75	17850	United Kingdom
536365	840296	KNITTED UNION FLA...	6	2010-12-01 08:26:00	3.39	17850	United Kingdom
536365	84029E	RED WOOLLY HOTTIE...	6	2010-12-01 08:26:00	3.39	17850	United Kingdom
536365	22752	SET 7 BABUSHKA NE...	2	2010-12-01 08:26:00	7.65	17850	United Kingdom
536365	21730	GLASS STAR FROSTE...	6	2010-12-01 08:26:00	4.25	17850	United Kingdom
536366	22633	HAND WARMER UNION...	6	2010-12-01 08:28:00	1.85	17850	United Kingdom

only showing top 8 rows

Произведем объединение 2-х файлов

Прочитаем 2-й файл

```
1 response2 = s3_client.get_object(  
2  
3     Bucket='lab-1-bucket-1',  
4     Key='kirill-gusev/retail/Online_Retail-2.csv' # Обратите внимание: Online, а не Online!  
5  
6 )  
7  
8 content2 = response2['Body'].read().decode('utf-8')  
9 df_pd2 = pd.read_csv(StringIO(content2))
```

```
/usr/local/lib/python3.12/dist-packages/urllib3/connectionpool.py:1097: InsecureRequestWarning: Unverified HTTPS request is being made to host 'minio-1.devops-playg  
warnings.warn(
```

Конвертируем в Spark

```
1 df_spark2 = spark.createDataFrame(df_pd2)
```

Произведем объединение

```
1 df_union = df_spark.union(df_spark2)  
2 print(f"Всего строк: {df_union.count()}")  
3 df_union.show(5)
```

Всего строк: 541909

InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
536365	85123A	WHITE HANGING HEA...	6	2010-12-01 08:26:00	2.55	17850	United Kingdom
536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	17850	United Kingdom
536365	84406B	CREAM CUPID HEART...	8	2010-12-01 08:26:00	2.75	17850	United Kingdom
536365	840296	KNITTED UNION FLA...	6	2010-12-01 08:26:00	3.39	17850	United Kingdom
536365	84029E	RED WOOLLY HOTTIE...	6	2010-12-01 08:26:00	3.39	17850	United Kingdom

only showing top 5 rows

Сохраним данные в Parquet

Для начало сохраним данные локально

```
1 local_path = "/content/retail_parquet"
2 df_union.write \
3     .mode("overwrite") \
4     .format("parquet") \
5     .save(local_path) ;      print(f"Данные сохранены локально: {local_path}")
```

Данные сохранены локально: /content/retail_parquet

Теперь произведем загрузку данных в S3

```
1 import os ; from boto3.client import Config
```

```
1 for root, dirs, files in os.walk(local_path):
2     for file in files:
3         local_file = os.path.join(root, file)
4         s3_key = f"kirill-gusev/retail_parquet/{file}"
5
6         with open(local_file, 'rb') as f:
7             s3_client.put_object(
8                 Bucket=BUCKET,
9                 Key=s3_key,
10                 Body=f
11             )
12         print(f"Загружен: {s3_key}")
```

```
/usr/local/lib/python3.12/dist-packages/urllib3/connectionpool.py:1097: InsecureRequestWarning: Unverified HTTPS request is being made to host 'minio-1.devops-playground.innopolis.university'.
warnings.warn(
Загружен: kirill-gusev/retail_parquet/part-00002-7caaaa95-fa4f-4ad5-aa28-178c0ed53b90-c000.snappy.parquet
/usr/local/lib/python3.12/dist-packages/urllib3/connectionpool.py:1097: InsecureRequestWarning: Unverified HTTPS request is being made to host 'minio-1.devops-playground.innopolis.university'.
warnings.warn(
Загружен: kirill-gusev/retail_parquet/.part-00000-7caaaa95-fa4f-4ad5-aa28-178c0ed53b90-c000.snappy.parquet.crc
/usr/local/lib/python3.12/dist-packages/urllib3/connectionpool.py:1097: InsecureRequestWarning: Unverified HTTPS request is being made to host 'minio-1.devops-playground.innopolis.university'.
warnings.warn(
Загружен: kirill-gusev/retail_parquet/part-00001-7caaaa95-fa4f-4ad5-aa28-178c0ed53b90-c000.snappy.parquet
/usr/local/lib/python3.12/dist-packages/urllib3/connectionpool.py:1097: InsecureRequestWarning: Unverified HTTPS request is being made to host 'minio-1.devops-playground.innopolis.university'.
warnings.warn(
Загружен: kirill-gusev/retail_parquet/.part-00002-7caaaa95-fa4f-4ad5-aa28-178c0ed53b90-c000.snappy.parquet.crc
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warnings.warn(
Загружен: kirill-gusev/retail_parquet/part-00003-7caaaa95-fa4f-4ad5-aa28-178c0ed53b90-c000.snappy.parquet
/usr/local/lib/python3.12/dist-packages/urllib3/connectionpool.py:1097: InsecureRequestWarning: Unverified HTTPS request is being made to host 'minio-1.devops-playground.innopolis.university'.
warnings.warn(
Загружен: kirill-gusev/retail_parquet/._SUCCESS.crc
/usr/local/lib/python3.12/dist-packages/urllib3/connectionpool.py:1097: InsecureRequestWarning: Unverified HTTPS request is being made to host 'minio-1.devops-playground.innopolis.university'.
warnings.warn(
Загружен: kirill-gusev/retail_parquet/.part-00003-7caaaa95-fa4f-4ad5-aa28-178c0ed53b90-c000.snappy.parquet.crc
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warnings.warn(
Загружен: kirill-gusev/retail_parquet/_SUCCESS
/usr/local/lib/python3.12/dist-packages/urllib3/connectionpool.py:1097: InsecureRequestWarning: Unverified HTTPS request is being made to host 'minio-1.devops-playground.innopolis.university'.
warnings.warn(
Загружен: kirill-gusev/retail_parquet/.part-00001-7caaaa95-fa4f-4ad5-aa28-178c0ed53b90-c000.snappy.parquet.crc
/usr/local/lib/python3.12/dist-packages/urllib3/connectionpool.py:1097: InsecureRequestWarning: Unverified HTTPS request is being made to host 'minio-1.devops-playground.innopolis.university'.
warnings.warn(
Загружен: kirill-gusev/retail_parquet/part-00000-7caaaa95-fa4f-4ad5-aa28-178c0ed53b90-c000.snappy.parquet
```


Добавляем колонку InvoiceMonth

Подгрузим дополнительные модули и библиотеки

```
1 import pyspark.sql.functions as F
2 import pandas as pd
3 from io import BytesIO
4 import pyarrow.parquet as pq
5 import pyarrow as pa
```

Получим список всех parquet файлов в папке

```
1 response = s3_client.list_objects_v2(
2     Bucket=BUCKET,
3     Prefix=f"{YOUR_NAME}/retail_parquet/"
4 )
```

```
/usr/local/lib/python3.12/dist-packages/urllib3/connectionpool.py:1097: InsecureRequestWarning: Unverified HTTPS request is being made to host 'minio-1.devops-playground.innopolis.university'.
warnings.warn()
```

Проверим содержимое

```
1 response = s3_client.list_objects_v2(
2     Bucket=BUCKET,
3     Prefix=f"{YOUR_NAME}/retail_parquet/"
4 )
5
6 if 'Contents' in response:
7     for obj in response['Contents']:
8         print(f" - {obj['Key']} ({obj['Size']} bytes)")
```

```
/usr/local/lib/python3.12/dist-packages/urllib3/connectionpool.py:1097: InsecureRequestWarning: Unverified HTTPS request is being made to host 'minio-1.devops-playground.innopolis.university'.
warnings.warn(
- kirill-gusev/retail_parquet/_SUCCESS.crc (8 bytes)
- kirill-gusev/retail_parquet/.part-00000-3aa0d93b-8cb9-4920-ae6a-6f3cfc0fe55f-c000.snappy.parquet.crc (6012 bytes)
- kirill-gusev/retail_parquet/.part-00000-7caaaa95-fa4f-4ad5-aa28-178c0ed53b90-c000.snappy.parquet.crc (6012 bytes)
- kirill-gusev/retail_parquet/.part-00001-3aa0d93b-8cb9-4920-ae6a-6f3cfc0fe55f-c000.snappy.parquet.crc (6284 bytes)
- kirill-gusev/retail_parquet/.part-00001-7caaaa95-fa4f-4ad5-aa28-178c0ed53b90-c000.snappy.parquet.crc (6284 bytes)
- kirill-gusev/retail_parquet/.part-00002-3aa0d93b-8cb9-4920-ae6a-6f3cfc0fe55f-c000.snappy.parquet.crc (8172 bytes)
- kirill-gusev/retail_parquet/.part-00002-7caaaa95-fa4f-4ad5-aa28-178c0ed53b90-c000.snappy.parquet.crc (8172 bytes)
- kirill-gusev/retail_parquet/.part-00003-3aa0d93b-8cb9-4920-ae6a-6f3cfc0fe55f-c000.snappy.parquet.crc (7888 bytes)
- kirill-gusev/retail_parquet/.part-00003-7caaaa95-fa4f-4ad5-aa28-178c0ed53b90-c000.snappy.parquet.crc (7888 bytes)
- kirill-gusev/retail_parquet/_SUCCESS (0 bytes)
- kirill-gusev/retail_parquet/part-00000-3aa0d93b-8cb9-4920-ae6a-6f3cfc0fe55f-c000.snappy.parquet (768272 bytes)
- kirill-gusev/retail_parquet/part-00000-7caaaa95-fa4f-4ad5-aa28-178c0ed53b90-c000.snappy.parquet (768272 bytes)
- kirill-gusev/retail_parquet/part-00001-3aa0d93b-8cb9-4920-ae6a-6f3cfc0fe55f-c000.snappy.parquet (802882 bytes)
- kirill-gusev/retail_parquet/part-00001-7caaaa95-fa4f-4ad5-aa28-178c0ed53b90-c000.snappy.parquet (802882 bytes)
- kirill-gusev/retail_parquet/part-00002-3aa0d93b-8cb9-4920-ae6a-6f3cfc0fe55f-c000.snappy.parquet (1044768 bytes)
- kirill-gusev/retail_parquet/part-00002-7caaaa95-fa4f-4ad5-aa28-178c0ed53b90-c000.snappy.parquet (1044768 bytes)
- kirill-gusev/retail_parquet/part-00003-3aa0d93b-8cb9-4920-ae6a-6f3cfc0fe55f-c000.snappy.parquet (1008401 bytes)
- kirill-gusev/retail_parquet/part-00003-7caaaa95-fa4f-4ad5-aa28-178c0ed53b90-c000.snappy.parquet (1008401 bytes)
```

```
1 response = s3_client.list_objects_v2(  
2     Bucket=BUCKET,  
3     Prefix=f"{YOUR_NAME}/retail_parquet/"  
4 )  
5  
6 all_data = []  
7  
8 if 'Contents' in response:  
9     for obj in response['Contents']:  
10         # Проверяем, что это parquet файл (по расширению)  
11         if obj['Key'].endswith('.parquet'):  
12             print(f"Чтение: {obj['Key']}")  
13  
14             # Скачиваем файл  
15             file_response = s3_client.get_object(Bucket=BUCKET, Key=obj['Key'])  
16  
17             # Читаем parquet  
18             df_pd = pd.read_parquet(BytesIO(file_response['Body'].read()))  
19  
20             print(f"{len(df_pd)} строк")  
21             all_data.append(df_pd)  
22         else:  
23             print(f"Пропускаем (не parquet): {obj['Key']}")  
24 else:  
25     print("Папка не найдена или пуста")  
26  
27 # Проверяем, что нашли файлы  
28 print(f"\nНайдено parquet файлов: {len(all_data)}")  
29  
30 if len(all_data) > 0:  
31     df_pd_all = pd.concat(all_data, ignore_index=True)  
32     print(f"Всего строк в объединенном файле: {len(df_pd_all)}")  
33     print("\nПервые 5 строк:")  
34     print(df_pd_all.head())  
35 else:  
36     print("Нет parquet файлов для обработки")
```

```

Пропускаем (не parquet): kirill-gusev/retail_parquet/.part-00000-7caeaa95-fa4f-4ad5-aa28-178c0ed53b90-c000.snappy.parquet.crc
Пропускаем (не parquet): kirill-gusev/retail_parquet/.part-00001-3aa0d93b-8cb9-4920-ae6a-6f3cfc0fe55f-c000.snappy.parquet.crc
Пропускаем (не parquet): kirill-gusev/retail_parquet/.part-00001-7caeaa95-fa4f-4ad5-aa28-178c0ed53b90-c000.snappy.parquet.crc
Пропускаем (не parquet): kirill-gusev/retail_parquet/.part-00002-3aa0d93b-8cb9-4920-ae6a-6f3cfc0fe55f-c000.snappy.parquet.crc
Пропускаем (не parquet): kirill-gusev/retail_parquet/.part-00002-7caeaa95-fa4f-4ad5-aa28-178c0ed53b90-c000.snappy.parquet.crc
Пропускаем (не parquet): kirill-gusev/retail_parquet/.part-00003-3aa0d93b-8cb9-4920-ae6a-6f3cfc0fe55f-c000.snappy.parquet.crc
Пропускаем (не parquet): kirill-gusev/retail_parquet/.part-00003-7caeaa95-fa4f-4ad5-aa28-178c0ed53b90-c000.snappy.parquet.crc
Пропускаем (не parquet): kirill-gusev/retail_parquet/.SUCCESS
Чтение: kirill-gusev/retail_parquet/part-00000-3aa0d93b-8cb9-4920-ae6a-6f3cfc0fe55f-c000.snappy.parquet
/usr/local/lib/python3.12/dist-packages/urllib3/connectionpool.py:1097: InsecureRequestWarning: Unverified HTTPS request is being made to host 'minio-1.devops-playground.innopolis.university'.
warnings.warn(
  → 114688 строк
Чтение: kirill-gusev/retail_parquet/part-00000-7caeaa95-fa4f-4ad5-aa28-178c0ed53b90-c000.snappy.parquet
/usr/local/lib/python3.12/dist-packages/urllib3/connectionpool.py:1097: InsecureRequestWarning: Unverified HTTPS request is being made to host 'minio-1.devops-playground.innopolis.university'.
warnings.warn(
  → 114688 строк
Чтение: kirill-gusev/retail_parquet/part-00001-3aa0d93b-8cb9-4920-ae6a-6f3cfc0fe55f-c000.snappy.parquet
/usr/local/lib/python3.12/dist-packages/urllib3/connectionpool.py:1097: InsecureRequestWarning: Unverified HTTPS request is being made to host 'minio-1.devops-playground.innopolis.university'.
warnings.warn(
  → 114818 строк
Чтение: kirill-gusev/retail_parquet/part-00001-7caeaa95-fa4f-4ad5-aa28-178c0ed53b90-c000.snappy.parquet
/usr/local/lib/python3.12/dist-packages/urllib3/connectionpool.py:1097: InsecureRequestWarning: Unverified HTTPS request is being made to host 'minio-1.devops-playground.innopolis.university'.
warnings.warn(
  → 114818 строк
Чтение: kirill-gusev/retail_parquet/part-00002-3aa0d93b-8cb9-4920-ae6a-6f3cfc0fe55f-c000.snappy.parquet
/usr/local/lib/python3.12/dist-packages/urllib3/connectionpool.py:1097: InsecureRequestWarning: Unverified HTTPS request is being made to host 'minio-1.devops-playground.innopolis.university'.
warnings.warn(
  → 156672 строк
Чтение: kirill-gusev/retail_parquet/part-00002-7caeaa95-fa4f-4ad5-aa28-178c0ed53b90-c000.snappy.parquet
/usr/local/lib/python3.12/dist-packages/urllib3/connectionpool.py:1097: InsecureRequestWarning: Unverified HTTPS request is being made to host 'minio-1.devops-playground.innopolis.university'.
warnings.warn(
  → 156672 строк
Чтение: kirill-gusev/retail_parquet/part-00003-3aa0d93b-8cb9-4920-ae6a-6f3cfc0fe55f-c000.snappy.parquet
/usr/local/lib/python3.12/dist-packages/urllib3/connectionpool.py:1097: InsecureRequestWarning: Unverified HTTPS request is being made to host 'minio-1.devops-playground.innopolis.university'.
warnings.warn(
  → 155731 строк
Чтение: kirill-gusev/retail_parquet/part-00003-7caeaa95-fa4f-4ad5-aa28-178c0ed53b90-c000.snappy.parquet
/usr/local/lib/python3.12/dist-packages/urllib3/connectionpool.py:1097: InsecureRequestWarning: Unverified HTTPS request is being made to host 'minio-1.devops-playground.innopolis.university'.
warnings.warn(
  → 155731 строк

```

Найдено parquet файлов: 8
Всего строк в объединенном файле: 1083818

Первые 5 строк:

	InvoiceNo	StockCode	Description	Quantity	\
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	
1	536365	71053	WHITE METAL LANTERN	6	
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	

	InvoiceDate	UnitPrice	CustomerID	Country
0	2010-12-01 08:26:00	2.55	17850	United Kingdom
1	2010-12-01 08:26:00	3.39	17850	United Kingdom
2	2010-12-01 08:26:00	2.75	17850	United Kingdom
3	2010-12-01 08:26:00	3.39	17850	United Kingdom
4	2010-12-01 08:26:00	3.39	17850	United Kingdom

Добавим колонку InvoiceMonth

```
1 # Преобразуем колонку InvoiceDate в datetime
2 df_pd_all['InvoiceDate'] = pd.to_datetime(df_pd_all['InvoiceDate'])
3
4 # Добавляем колонку с первым днем месяца
5 df_pd_all['InvoiceMonth'] = df_pd_all['InvoiceDate'].dt.to_period('M').dt.start_time
6
7 print("Результат - первые 10 строк с новой колонкой:")
8 print(df_pd_all[['InvoiceDate', 'InvoiceMonth']].head(10))
9
10 print("\nУникальные месяцы в данных:")
11 print(df_pd_all['InvoiceMonth'].unique())
```

Результат - первые 10 строк с новой колонкой:

	InvoiceDate	InvoiceMonth
0	2010-12-01 08:26:00	2010-12-01
1	2010-12-01 08:26:00	2010-12-01
2	2010-12-01 08:26:00	2010-12-01
3	2010-12-01 08:26:00	2010-12-01
4	2010-12-01 08:26:00	2010-12-01
5	2010-12-01 08:26:00	2010-12-01
6	2010-12-01 08:26:00	2010-12-01
7	2010-12-01 08:28:00	2010-12-01
8	2010-12-01 08:28:00	2010-12-01
9	2010-12-01 08:34:00	2010-12-01

Уникальные месяцы в данных:

```
<DatetimeArray>
['2010-12-01 00:00:00', '2011-01-01 00:00:00', '2011-02-01 00:00:00',
 '2011-03-01 00:00:00', '2011-04-01 00:00:00', '2011-05-01 00:00:00',
 '2011-06-01 00:00:00', '2011-07-01 00:00:00', '2011-08-01 00:00:00',
 '2011-09-01 00:00:00', '2011-10-01 00:00:00', '2011-11-01 00:00:00',
 '2011-12-01 00:00:00']
Length: 13, dtype: datetime64[ns]
```

Сохраним данные с новой колонкой

```
1 import pyarrow as pa
2 import pyarrow.parquet as pq
3
4 # Сохраняем в parquet
5 buffer = BytesIO()
6 df_pd_all.to_parquet(buffer, index=False, engine='pyarrow')
7 buffer.seek(0)
8
9 # Загружаем в S3
10 month_path = f"{YOUR_NAME}/retail_with_month/data.parquet"
11 s3_client.put_object(
12     Bucket=BUCKET,
13     Key=month_path,
14     Body=buffer.getvalue()
15 )
16
17 print(f"Данные с колонкой InvoiceMonth сохранены в: {month_path}")
18
19 response = s3_client.get_object(Bucket=BUCKET, Key=month_path)
20 df_check = pd.read_parquet(BytesIO(response['Body'].read()))
21
22 print(f"Проверка: {len(df_check)} строк сохранено")
23 print(df_check[['InvoiceDate', 'InvoiceMonth']].head())
```

```
/usr/local/lib/python3.12/dist-packages/urllib3/connectionpool.py:1097: InsecureRequestWarning: Unverified HTTPS request is being made to host 'minio-1.devops-playground.innopolis.university'.
warnings.warn(
Данные с колонкой InvoiceMonth сохранены в: kirill-gusev/retail_with_month/data.parquet
/usr/local/lib/python3.12/dist-packages/urllib3/connectionpool.py:1097: InsecureRequestWarning: Unverified HTTPS request is being made to host 'minio-1.devops-playground.innopolis.university'.
warnings.warn(
Проверка: 1083818 строк сохранено
   InvoiceDate InvoiceMonth
0 2010-12-01 08:26:00    2010-12-01
1 2010-12-01 08:26:00    2010-12-01
2 2010-12-01 08:26:00    2010-12-01
3 2010-12-01 08:26:00    2010-12-01
4 2010-12-01 08:26:00    2010-12-01
```

Запрос без партиционирования (через pandas)

```
1  # Уникальные клиенты за декабрь 2010
2
3  dec_2010 = df_pd_all[df_pd_all['InvoiceMonth'] == '2010-12-01']
4  customers_dec = dec_2010['CustomerID'].dropna().unique()
5
6  print(f"Уникальные клиенты за декабрь 2010: {len(customers_dec)}")
7  print("Первые 10:")
8  print(customers_dec[:10])
9
10 # Группировка по клиентам (альтернативный запрос)
11
12 customer_stats = dec_2010.groupby('CustomerID').agg({
13     'InvoiceNo': 'nunique',
14     'Quantity': 'sum',
15     'UnitPrice': 'mean'
16 }).rename(columns={
17     'InvoiceNo': 'orders_count',
18     'Quantity': 'total_items',
19     'UnitPrice': 'avg_price'
20 }).reset_index()
21
22 print("Статистика по клиентам за декабрь 2010:")
23 print(customer_stats.head(10))
```

Уникальные клиенты за декабрь 2010: 949

Первые 10:

[17850 13047 12583 13748 15100 15291 14688 17809 15311 14527]

Статистика по клиентам за декабрь 2010:

	CustomerID	orders_count	total_items	avg_price
0	12347	1	638	2.890000
1	12348	1	2508	2.917647
2	12370	2	1936	2.894286
3	12377	1	1208	2.106279
4	12383	1	1508	1.325135
5	12386	1	428	2.726250
6	12395	2	1506	3.390000
7	12417	1	120	7.003636
8	12423	1	376	2.636875
9	12427	2	140	5.226923

Партиционирование по месяцу

```
1 import os
2
3 # Группируем данные по месяцам
4 df_pd_all['InvoiceMonth'] = pd.to_datetime(df_pd_all['InvoiceMonth']).dt.strftime('%Y-%m-%d')
5
6 for month, group in df_pd_all.groupby('InvoiceMonth'):
7     print(f"Сохраняем данные за {month}...")
8
9     # Сохраняем в буфер
10    buffer = BytesIO()
11    group.to_parquet(buffer, index=False, engine='pyarrow')
12    buffer.seek(0)
13
14    # Загружаем в S3 в соответствующую папку
15    month_path = f"{YOUR_NAME}/retail_partitioned/InvoiceMonth={month}/data.parquet"
16    s3_client.put_object(
17        Bucket=BUCKET,
18        Key=month_path,
19        Body=buffer.getvalue()
20    )
21    print(f"{len(group)} строк сохранено в {month_path}")
22
23 print("Все данные разбиты по месяцам и сохранены")
```

```
Сохраняем данные за 2010-12-01...
/usr/local/lib/python3.12/dist-packages/urllib3/connectionpool.py:1097: InsecureRequestWarning: Unverified HTTPS request is being made to host 'minio-1.devops-playground.innopolis.university'.
  warnings.warn(
84962 строк сохранено в kirill-gusev/retail_partitioned/InvoiceMonth=2010-12-01/data.parquet
Сохраняем данные за 2011-01-01...
/usr/local/lib/python3.12/dist-packages/urllib3/connectionpool.py:1097: InsecureRequestWarning: Unverified HTTPS request is being made to host 'minio-1.devops-playground.innopolis.university'.
  warnings.warn(
70294 строк сохранено в kirill-gusev/retail_partitioned/InvoiceMonth=2011-01-01/data.parquet
Сохраняем данные за 2011-02-01...
/usr/local/lib/python3.12/dist-packages/urllib3/connectionpool.py:1097: InsecureRequestWarning: Unverified HTTPS request is being made to host 'minio-1.devops-playground.innopolis.university'.
  warnings.warn(
55414 строк сохранено в kirill-gusev/retail_partitioned/InvoiceMonth=2011-02-01/data.parquet
Сохраняем данные за 2011-03-01...
/usr/local/lib/python3.12/dist-packages/urllib3/connectionpool.py:1097: InsecureRequestWarning: Unverified HTTPS request is being made to host 'minio-1.devops-playground.innopolis.university'.
  warnings.warn(
73496 строк сохранено в kirill-gusev/retail_partitioned/InvoiceMonth=2011-03-01/data.parquet
Сохраняем данные за 2011-04-01...
/usr/local/lib/python3.12/dist-packages/urllib3/connectionpool.py:1097: InsecureRequestWarning: Unverified HTTPS request is being made to host 'minio-1.devops-playground.innopolis.university'.
  warnings.warn(
59832 строк сохранено в kirill-gusev/retail_partitioned/InvoiceMonth=2011-04-01/data.parquet
Сохраняем данные за 2011-05-01...
/usr/local/lib/python3.12/dist-packages/urllib3/connectionpool.py:1097: InsecureRequestWarning: Unverified HTTPS request is being made to host 'minio-1.devops-playground.innopolis.university'.
  warnings.warn(
74060 строк сохранено в kirill-gusev/retail_partitioned/InvoiceMonth=2011-05-01/data.parquet
Сохраняем данные за 2011-06-01...
/usr/local/lib/python3.12/dist-packages/urllib3/connectionpool.py:1097: InsecureRequestWarning: Unverified HTTPS request is being made to host 'minio-1.devops-playground.innopolis.university'.
  warnings.warn(
73748 строк сохранено в kirill-gusev/retail_partitioned/InvoiceMonth=2011-06-01/data.parquet
Сохраняем данные за 2011-07-01...
/usr/local/lib/python3.12/dist-packages/urllib3/connectionpool.py:1097: InsecureRequestWarning: Unverified HTTPS request is being made to host 'minio-1.devops-playground.innopolis.university'.
  warnings.warn(
79036 строк сохранено в kirill-gusev/retail_partitioned/InvoiceMonth=2011-07-01/data.parquet
Сохраняем данные за 2011-08-01...
/usr/local/lib/python3.12/dist-packages/urllib3/connectionpool.py:1097: InsecureRequestWarning: Unverified HTTPS request is being made to host 'minio-1.devops-playground.innopolis.university'.
  warnings.warn(
70568 строк сохранено в kirill-gusev/retail_partitioned/InvoiceMonth=2011-08-01/data.parquet
Сохраняем данные за 2011-09-01...
/usr/local/lib/python3.12/dist-packages/urllib3/connectionpool.py:1097: InsecureRequestWarning: Unverified HTTPS request is being made to host 'minio-1.devops-playground.innopolis.university'.
  warnings.warn(
100452 строк сохранено в kirill-gusev/retail_partitioned/InvoiceMonth=2011-09-01/data.parquet
Сохраняем данные за 2011-10-01...
/usr/local/lib/python3.12/dist-packages/urllib3/connectionpool.py:1097: InsecureRequestWarning: Unverified HTTPS request is being made to host 'minio-1.devops-playground.innopolis.university'.
  warnings.warn(
121484 строк сохранено в kirill-gusev/retail_partitioned/InvoiceMonth=2011-10-01/data.parquet
Сохраняем данные за 2011-11-01...
/usr/local/lib/python3.12/dist-packages/urllib3/connectionpool.py:1097: InsecureRequestWarning: Unverified HTTPS request is being made to host 'minio-1.devops-playground.innopolis.university'.
  warnings.warn(
169422 строк сохранено в kirill-gusev/retail_partitioned/InvoiceMonth=2011-11-01/data.parquet
Сохраняем данные за 2011-12-01...
/usr/local/lib/python3.12/dist-packages/urllib3/connectionpool.py:1097: InsecureRequestWarning: Unverified HTTPS request is being made to host 'minio-1.devops-playground.innopolis.university'.
  warnings.warn(
51050 строк сохранено в kirill-gusev/retail_partitioned/InvoiceMonth=2011-12-01/data.parquet
Все данные разбиты по месяцам и сохранены
```


Проверим как прошло партиционирование

```
1 # Проверяем структуру папок
2 response = s3_client.list_objects_v2(
3     Bucket=BUCKET,
4     Prefix=f"{YOUR_NAME}/retail_partitioned/"
5 )
6
7 folders = set()
8 for obj in response.get('Contents', []):
9     parts = obj['Key'].split('/')
10    if len(parts) > 2 and 'InvoiceMonth=' in parts[1]:
11        folders.add(parts[1])
12
13 print("Созданные партиции: \n")
14 for folder in sorted(folders):
15     print(f" - {folder}")
```

Созданные партиции:

/usr/local/lib/python3.12/dist-packages/urllib3/connectionpool.py:1097: InsecureRequestWarning: Unverified HTTPS request is being made to host 'minio-1.devops-playground.innopolis.university'. ,
warnings.warn(



lab-1-bucket-1

Created on: **Mon, Apr 28 2025 12:04:19 (GMT+3)** Access: **PRIVATE** 969.4 MiB - 292 Object



lab-1-bucket-1 / kirill-gusev



▲ Name

Last Modified



📁 retail_parquet



📁 retail_partitioned



📁 retail_with_month



📁 retail



lab-1-bucket-1

Created on: Mon, Apr 28 2025 12:04:19 (GMT+3) Access: PRIVATE 969.4 MiB - 292 Object



lab-1-bucket-1 / kirill-gusev / retail_parquet

☐	▲ Name	Last Modified
☐	 _SUCCESS	Mon, Mar 16 2026 22:24 (GMT+3)
☐	 ._SUCCESS.crc	Mon, Mar 16 2026 22:24 (GMT+3)
☐	 .part-00000-3aa0d93b-8cb9-4920-ae6a-6f3cfc0fe55f-c000.snappy.parquet.crc	Mon, Mar 16 2026 20:15 (GMT+3)
☐	 .part-00000-7caeea95-fa4f-4ad5-aa28-178c0ed53b90-c000.snappy.parquet.crc	Mon, Mar 16 2026 22:24 (GMT+3)
☐	 .part-00001-3aa0d93b-8cb9-4920-ae6a-6f3cfc0fe55f-c000.snappy.parquet.crc	Mon, Mar 16 2026 20:15 (GMT+3)
☐	 .part-00001-7caeea95-fa4f-4ad5-aa28-178c0ed53b90-c000.snappy.parquet.crc	Mon, Mar 16 2026 22:24 (GMT+3)
☐	 .part-00002-3aa0d93b-8cb9-4920-ae6a-6f3cfc0fe55f-c000.snappy.parquet.crc	Mon, Mar 16 2026 20:15 (GMT+3)
☐	 .part-00002-7caeea95-fa4f-4ad5-aa28-178c0ed53b90-c000.snappy.parquet.crc	Mon, Mar 16 2026 22:24 (GMT+3)
☐	 .part-00003-3aa0d93b-8cb9-4920-ae6a-6f3cfc0fe55f-c000.snappy.parquet.crc	Mon, Mar 16 2026 20:15 (GMT+3)
☐	 .part-00003-7caeea95-fa4f-4ad5-aa28-178c0ed53b90-c000.snappy.parquet.crc	Mon, Mar 16 2026 22:24 (GMT+3)
☐	 part-00000-3aa0d93b-8cb9-4920-ae6a-6f3cfc0fe55f-c000.snappy.parquet	Mon, Mar 16 2026 20:15 (GMT+3)



lab-1-bucket-1

Created on: **Mon, Apr 28 2025 12:04:19 (GMT+3)** Access: **PRIVATE** 969.4 MiB - 292 Object



lab-1-bucket-1 / kirill-gusev / retail_partitioned



▲ Name

Last Modified



📁 InvoiceMonth=2010-12-01



📁 InvoiceMonth=2011-01-01



📁 InvoiceMonth=2011-02-01



📁 InvoiceMonth=2011-03-01



📁 InvoiceMonth=2011-04-01



📁 InvoiceMonth=2011-05-01



📁 InvoiceMonth=2011-06-01



📁 InvoiceMonth=2011-07-01



📁 InvoiceMonth=2011-08-01



📁 InvoiceMonth=2011-09-01



📁 InvoiceMonth=2011-10-01



lab-1-bucket-1

Created on: Mon, Apr 28 2025 12:04:19 (GMT+3) Access: **PRIVATE** 969.4 MiB - 292 Object



lab-1-bucket-1 / kirill-gusev / retail_with_month



▲ Name

Last Modified



📄 data.parquet

Mon, Mar 16 2026 22:24 (GMT+3)